Dr. Sadullah GONCU

Research Assistant / Ph.D.
Department of Civil Engineering, Fatih Sultan Mehmet Vakif University
Istanbul, Türkiye

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Research Interests

- Transportation Engineering, Traffic Flow Theories, Intelligent Transportation Systems.
- Disaster Management, Logistics, Smart Logistics
- Control Theory
- Sustainable Transportation Planning

Employment

Research Assistant 2017–Ongoing
Department of Civil Engineering, Fatih Sultan Mehmet Vakif University

- Teaching Assistant duties on Transportation Engineering, Railway Engineering, Transportation Design, Civil Engineering Design, and Computer Programming courses.
- Administrative duties within the Internship Commission, Accreditation Board Commission

Education

Ph.D. in Transportation Engineering 2020–16.01.2026
Istanbul Technical University, Istanbul, Türkiye
Dissertation: *A Modeling and Control Framework for Mixed-Traffic Freeway Operations*
Supervisor(s): Prof. Dr. Hilmi Berk CELIKOGLU, Dr. Mehmet Ali SILGU

M.Sc. in Transportation Engineering 2017–16.01.2020
Yildiz Technical University, Istanbul, Türkiye
Thesis: *GPS-based Incident Detection Algorithm for Istanbul's Metrobus System*
Supervisor(s): Prof. Dr. İsmail ŞAHİN

B.Sc. in Civil Engineering 2013–2017
Fatih Sultan Mehmet Vakif University, Istanbul, Türkiye

Publications

- **Goncu, S.**, Silgu, M.A., Celikoglu, H.B. (Submitted at August 2025). Robust Gain-Scheduled Continuous-Time Linear Quadratic Regulator for Mixed-Traffic Freeways: A

Multi-Class Cell Transmission Model Approach. *IEEE Transactions on Intelligent Transportation Systems,* (Accepted, awaiting for proofreading, Scopus Q1, Wos Q1, Top 10% Journal).

- Akova, H., **Goncu, S.**, Silgu, M.A., Celikoglu, H.B. (2025). Traffic Congestion in Routing Problems: Insights from Autonomous Mobility-on-Demand and Shared Ride Systems. Arabian Journal for Science and Engineering, *Arabian Journal for Science and Engineering,* 1-13 (Early Access) (Scopus Q1 , WoS Q2, SCI-E).
- **Goncu, S.**, Silgu, M. A. (2025). Impacts of Car-Following Models on Simulation-Based Safety Evaluation of Freeways. *Arabian Journal for Science and Engineering,* 1-15 (Early Access) (Scopus Q1 , WoS Q2, SCI-E).
- Civelek, D. O., Goncu, B., **Goncu, S.**, & Okyar, A. (2025). Effects of everolimus on circadian gene expression and cell fate in synchronized Caco-2 cells.*Experimental Cell Research,*, 114810.(Scopus Q2 , WoS Q2, SCI-E)
- **Goncu, S.**, & Sahin, I. (2024). GPS-based incident detection algorithm for two-lane bus rapid transit systems: case study of Istanbul Metrobus. *Public Transport,* (Scopus Q1, ESCI) 16(3), 803-829.
- Akyol, G., **Göncü, S.**, & Silgu, M. A. (2024). Multi-objective optimization framework for Trade-Off among pedestrian delays and vehicular emissions at Signal-Controlled intersections. *Arabian Journal for Science and Engineering,* 49(10), 14117-14130. (Scopus Q2 , WoS Q2, SCI-E)
- **Goncu, S.**, Silgu, M. A., Akova, H., & Celikoglu, H. B. (2025). Exploring effect of CACC vehicle types on freeway traffic flow: Case study on a Bosphorus Strait. *Transportation Research Procedia,* 86, 467-474 (Scopus Q3).
- Akyol, G., Silgu, M. A., **Goncu, S.**, & Celikoglu, H. B. (2024). A bi-objective traffic signal optimization model for mixed traffic concerning pedestrian delays.*Transportation research procedia,* 78, 182-189 (Scopus Q3).

Conference Proceedings, Book Chapters

- **Goncu, S.** (2025, September). Reinforcement Learning-Based Freeway Traffic Control Concerning Emissions. *EWGT2025 Conference Proceedings,* To be published in Transportation Research Procedia (Scopus Q3) (1st Quarter 2026).
- **Goncu, S.**, Silgu, M. A., Akova, H., & Celikoglu, H. B. (2025). Exploring effect of CACC vehicle types on freeway traffic flow: Case study on a Bosphorus Strait. *EWGT2024 Conference Proceedings,*
- Sağlam, M. M., Önal, Ö. Y., Keskin, M., Kuş, Z., **Göncü, S.**, Çakır, Ö., & Yıldırım, D. (2025, June). Evaluation of Concrete Performance of Risky Buildings in Earthquakes using Machine Learning. *2025 33rd Signal Processing and Communications Applications Conference (SIU) (pp. 1-4),* IEEE.
- Akyol, G., Silgu, M. A., **Goncu, S.**, & Celikoglu, H. B. (2024). A bi-objective traffic signal optimization model for mixed traffic concerning pedestrian delays.*EWGT2023 Conference Proceedings.*
- Silgu, M.A., Goksad, I.E., Hulagu, S., Akti, S., Akova, H., Akyol, G., **Goncu, S.**, Goksu, G., Celikoglu, H.B. (2022). A Methodology to Consider Explicitly Emissions in Dynamic User Equilibrium Assignment. *Computer Aided Systems Theory – EUROCAST 2022, Lecture Notes in Computer Science, vol 13789,* Springer, Cham, pp. 512-523.
- **Goncu, S.**, Ali Silgu, M., Goksad Erdagı, I., Berk Celikoglu, H. (2022). Sensitivity Analysis for a Cooperative Adaptive Cruise Control Car Following Model: Preliminary Findings. *Computer Aided Systems Theory – EUROCAST 2022, Lecture Notes in Computer Science, vol 13789,* Springer, Cham, pp. 524-535.
- **Goncu, S.**, Silgu, M. A., & Celikoglu, H. B. (2022, June). Evaluation of vehicle assign-

ment algorithms for autonomous mobility on demand. *2022 IEEE Intelligent Vehicles Symposium (IV) (pp. 702-707),* IEEE.

- **Goncu, S.**, Erdagi, I. G., Silgu, M. A., & Celikoglu, H. B. (2022, June). Analysis on effects of driving behavior on freeway traffic flow: A comparative evaluation of two driver profiles using two car-following models. *2022 IEEE Intelligent Vehicles Symposium (IV) (pp. 688-693),* IEEE.
- **Göncü, S.**, Silgu, M.A., Çelikoğlu H.B. (2024). Doğal Afetler ve Şehirlerin Tahliyesi (Disasters and Evacuations). *Dünya Şehircilik Günü 48. Kolokyumu , TMMOB Şehir Plancıları Odası,* 7-9 Kasım 2024, Antalya, Türkiye.
- **Göncü, S.**, Şahin, İ. (2019). İstanbul Metrobüs Koridorundaki Trafik Akımlarının GPS Verilerinden Yararlanılarak İncelenmesi. *TMMOB İnşaat Mühendisleri Odası 13. Ulaştırma Kongresi,* 10-12 Ekim 2019, Erzurum, Türkiye.

Projects

- **Research Fellow** at "Effects of Cooperative Vehicle Dynamics in Traffic Flow Modeling: Simulation and Control," the Scientific and Technological Research Council of Turkey (TUBITAK-1001) National Research Project No. 120M576, PI: Hilmi Berk Celikoglu, 2020-2023. As an extension of his doctoral dissertation, **Dr. Goncu** was responsible for the design and implementation of the data collection process, the development of simulation models, the integration of car-following models into both commercial and open-source traffic microsimulation softwares, and the evaluation of the performance of control algorithms for mainstream and on-ramps.
- **Research Fellow** at Center of Excellence Project No. 22AG018, titled "Sustainable Advanced Vehicle Technologies," funded by the Scientific and Technological Research Council of Turkey (TUBITAK-1004), 15.05.2023-15.04.2024. **Dr. Goncu** was responsible for environmentally friendly eco-routing solutions for electric vehicles.
- **Research Fellow** at "Improving routing solutions for on-demand mobility systems through simulation: an integrated modeling and optimization framework and its evaluation through the case of a low emission zone, The Historical Peninsula, İstanbul" the Scientific and Technological Research Council of Turkey (TUBITAK-1001) National Research Project No. 124M395, PI: Hilmi Berk Celikoglu, 24.11.2024-16.01.2026. **Dr. Goncu** was heavily involved in the development and writing of the project proposal, and the first development report period. His duties and contribution of the project consists of development of simulation models, the formulation and evaluation of policy scenarios, the design and execution of data collection processes, and the development of the optimization framework used to assess routing solutions for on-demand mobility systems under low-emission zone constraints in the Historical Peninsula of Istanbul. **Dr. Goncu** is expected to join to the project as a researcher in the coming months.

Grants, Memberships, Fellowships & Awards

- 2020-ongoing Member of IEEE ITS Society
- 2020-ongoing Member of ITS Research Lab at Istanbul Technical University
- 2020-ongoing Member of Union of Chambers of Turkish Engineers and Architects
- 2023-2024 Academic Performance Award by Fatih Sultan Mehmet Vakif University Rectorate
- 2024-2025 Academic Performance Award by Fatih Sultan Mehmet Vakif University Rectorate

Reviewing Activities

- IEEE Transactions on Intelligent Transportation Systems
- Scientific Reports
- Public Transport (Springer)
- Arabian Journal for Science and Engineering
- Simulation Modelling Practice and Theory
- Transportation Research Record
- Transportation Letters: The International Journal of Transportation Research
- Discover Vehicles
- IEEE Intelligent Vehicles Symposium
- IEEE International Conference on Intelligent Transportation Systems
- TEKNOFEST Intelligent Transportation Systems Jury

Advanced Training, Professional Development

- TU Delft University, TRAIL (The Research School for TRAnsport, Infrastructure and Logistics), Ph.D. Summer School on Automated Driving, Delft, Netherlands, 20222
- European University (EELISA), EELISA on the Move community, Workshop "Low Emission Zones: Challenges and Opportunities", Istanbul, Türkiye, 2023 - Local Comittee Member
- European University (EELISA), EELISA on the Move community, Seminar and Workshop "Urban Mobility Solutions For Liveable Cities", Budapest, Hungary, 2023 - Attendee

Skills

Programming: Python, MATLAB, Gurobi, C++, SUMO, AIMSUN
Languages: Turkish (native), English (fluent) **YÖKDİL Exam:** 90/100 (29.08.2021)